

A Comparative Analysis of Time-Delay Strategies for Phased-Array Parametric Loudspeakers

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Abstract—In conventional linear-acoustics-based phased arrays, beam control is achieved through two degrees of freedom: applying time delays to set the phase and applying weights to control the amplitude. However, parametric array loudspeakers (PALs) generate audible sound through the self-demodulation of modulated ultrasonic signals, which introduces a previously unexplored degree of freedom: delays can be applied either before modulation (on the audio or ultrasonic signal) or after modulation (on the carrier or sideband). In this work, we take beam steering as an example to systematically compare five time-delay strategies using a k -space model of nonlinear propagation on a representative PAL array. A counterintuitive result is that delaying only the audio signal, though common in conventional loudspeaker arrays, cannot steer the audio beam effectively. By contrast, delaying the entire modulated signal achieves the optimal steering by aligning both the carrier and sideband, but its implementation demands sampling rates at the ultrasonic level. An interesting finding is that delaying only the ultrasonic signal achieves performance close to delaying the entire modulated signal case while requiring only single-frequency phase shifts, making it easier to implement in practice. The proposed framework unifies these observations within the quasi-linear model and clarifies how delay placement shapes the effective virtual source distribution. Experimental measurements further validate the key simulation findings. Together, these results establish delay placement as a new control dimension in phased-array PALs and provide a systematic foundation for future design and analysis.

Index Terms—Phased array, parametric array loudspeaker, time-delay strategy, beam steering, k -space method.

I. INTRODUCTION

PHASED arrays are widely employed for beam control in acoustics [1], [2], [3], [4], [5]. Beyond conventional

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applications, this principle has also been extended to PALs, where the nonlinear self-demodulation of ultrasonic signals transfers the high directivity of ultrasonic phased arrays into the audible range, thereby enabling compact directional audio systems [6], [7], [8], [9]. Early studies on phased-array PALs can be traced back to the 1980 s. Yoneyama et al. introduced the “audio spotlight” as the first directional PAL [10], followed by Gan’s digital beam steering method and his systematic review of phased-array strategies [7], [11]. Shi’s dissertation further investigated beamforming and grating-lobe suppression [12], and Fan et al. presented a least-squares optimization method for grating-lobe suppression [13]. Recently, optimization-based beamforming methods such as minimum-variance distortionless response (MVDR) have also been introduced to enhance steering performance in multi-channel PAL arrays [14]. Zhong et al. applied sparse random arrays to suppress sidelobes [15], Zhuang et al. developed a spherical-wave-expansion model incorporating inter-element nonlinear coupling [16], and Zhong et al. later engineered ultrasonic phase distributions to create self-bending beams that can form localized audible enclaves [17]. Together, these works establish phased-array PALs as a powerful framework for directional sound control. However, unlike conventional loudspeaker arrays, which perform beamforming directly in the audio band, phased-array PALs offer an additional degree of freedom: time delays can be applied at different stages of the modulation chain. In this work, we focus exclusively on time-delay control while keeping element amplitudes identical, following the classical phased-array formulation. This allows the structural impact of delay placement to be isolated. Introducing amplitude weighting would extend the problem to a more general multichannel processing framework, whereas the present study concentrates on phase-only control.

A straightforward idea is to apply time delays directly to the audio signal (audio-only delay, AD). However, since the ultrasonic waves do not inherit the required spatial phase profile, this approach is generally ineffective for steering the difference-frequency field and has therefore seen little use in practice. To overcome this limitation, some studies have applied time delays to the fully modulated ultrasonic signals (full-band delay, FD) [11], [12], [18], [19], [20], [21]. By introducing time delays to the complete modulated waveform, FD enforces consistent phase alignment of both carrier and sideband components across the array, thereby enabling accurate beam steering with precise control of the steering angle. In practical implementations, such accuracy is typically achieved using fractional delay filters.

However, realizing wideband fractional delays at ultrasonic sampling rates requires high-order filtering and increased computational resources, which may impose substantial implementation cost in real-time systems [22]. More recently, ultrasound-only delay (UD) has also been explored [22], [23], [24], [25]. Unlike FD, which delays the fully modulated signal, UD applies phase shifts only to the ultrasonic carrier prior to modulation. This structural difference simplifies implementation: the required operation reduces to per-channel phase adjustments, enabling efficient Field-Programmable Gate Array (FPGA) realization using hardware multipliers for real-time modulation. Although UD has been shown to produce steering effects in practice, its physical relationship to FD is not self-evident, and the mechanism by which it shapes the audible difference-frequency field remains insufficiently understood. In particular, despite the absence of explicit audio delays, the resulting difference-frequency field still exhibits a clear and measurable steering behavior, which calls for systematic theoretical analysis.

From a theoretical perspective, additional strategies such as applying time delays only to the carrier (carrier-only delay, CD) or to the sideband (sideband-only delay, SD) may also be considered, but these remain unexplored. In particular, applying delays to both CD and SD, or simultaneously to AD and UD, is mathematically equivalent to FD. Taken together, the existing approaches (AD, FD, UD) highlight different trade-offs among simplicity, performance, and computational complexity, yet their mechanisms and interrelations have not been systematically explained. This gap motivates the present work. Taking the beam steering as an example, we extend the scope of delay-placement strategies to include CD and SD, and establish a unified framework to interpret the performance of all five strategies. The proposed framework combines theoretical analysis, numerical simulations, and experimental validation, providing new insights into how delay placement shapes beam steering in phased-array PALs.

To ensure both accuracy and efficiency in evaluating these strategies, we adopt a recently developed k -space modeling method for PALs [26]. The key principle of the k -space approach is to transform the Rayleigh integral and the nonlinear volume convolution into spectral multiplications in the spatial-frequency domain. Ultrasonic primary waves are computed using the angular-spectrum approach (ASA), while the audible difference-frequency field is obtained by multiplying the virtual-source spectrum with the Green's function in k -space, followed by an inverse Fast Fourier Transform (FFT) [27], [28]. This formulation avoids the prohibitive cost of direct integration [8], yet faithfully preserves free-space propagation and nonlinear self-demodulation. By further incorporating discrete transducer sampling into the k -space formulation, we extend the framework into a practical k -space phased-array model that can directly simulate multi-channel PALs with arbitrary element layouts. This unified description provides both a continuous-source benchmark and a discrete-array tool, enabling systematic and physically consistent comparisons of the five time-delay strategies. In addition, experimental measurements are conducted with a prototype PAL system to validate the key conclusions drawn from the theoretical analysis and simulation results.

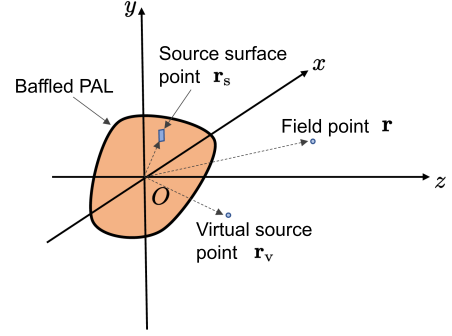


Fig. 1. Schematic of the PAL geometry and coordinate system.

The remainder of this work is organized as follows. Section II introduces the theoretical framework, including the quasi-linear solution, the k -space method, and five delay-placement strategies. Section III presents simulation results for an ideal continuous source distribution, which serve as a theoretical benchmark. Section IV describes the experimental study using a discrete PAL array, together with corresponding simulations for direct comparison. Finally, Section V concludes the work and discusses potential directions for future work.

II. THEORY

A. Quasi-Linear Model of PALs

As illustrated in Fig. 1, the PAL with a continuous aperture profile is considered. The array is mounted on a rigid baffle lying in the Oxy plane, and a coordinate system $Oxyz$ is defined with the z -axis perpendicular to the aperture. A point on the radiating surface is denoted as $\mathbf{r}_s = (x_s, y_s, 0)$, a general field location by $\mathbf{r} = (x, y, z)$, and a representative virtual audio source point by $\mathbf{r}_v = (x_v, y_v, z_v)$. The ambient density and sound speed are ρ_0 and c_0 , respectively, while β denotes the coefficient of nonlinearity of air.

The PAL emits two ultrasonic components at frequencies $f_1 < f_2$ with angular frequencies $\omega_i = 2\pi f_i$, $i = 1, 2$. The normal surface vibration velocity can be written as

$$v_z(\mathbf{r}_s, t) = v_{1,z}(\mathbf{r}_s)e^{-i\omega_1 t} + v_{2,z}(\mathbf{r}_s)e^{-i\omega_2 t}, \quad (1)$$

where $v_{i,z}(\mathbf{r}_s)$ denotes the on-surface velocity distribution at frequency f_i . According to the Rayleigh integral [8], the ultrasonic pressure at a virtual source point $\mathbf{r}_v$ is expressed as

$$p_i(\mathbf{r}_v) = -2i\rho_0\omega_i \iint_S v_{i,z}(\mathbf{r}_s)g(\mathbf{r}_v, \mathbf{r}_s; \omega_i)d^2\mathbf{r}_s, \quad i = 1, 2, \quad (2)$$

with the Green's function

$$g(\mathbf{r}_v, \mathbf{r}_s; \omega_i) = \frac{e^{ik_i\|\mathbf{r}_v - \mathbf{r}_s\|}}{4\pi\|\mathbf{r}_v - \mathbf{r}_s\|}, \quad k_i = \frac{\omega_i}{c_0} + i\alpha(\omega_i), \quad (3)$$

where $\alpha(\omega_i)$ is the absorption coefficient of air at the frequency f_i .

Following the quasi-linear solution, the audible sound field can be regarded as the superposition of secondary sources distributed in the volume. The source density is given by

$$q(\mathbf{r}_v) = \frac{\beta\omega_a}{i\rho_0^2 c_0^4} p_1^*(\mathbf{r}_v) p_2(\mathbf{r}_v), \quad \omega_a = 2\pi f_a, \quad f_a = f_2 - f_1, \quad (4)$$

where the superscript “*” denotes the complex conjugate. Here ω_1 and ω_2 denote the two ultrasonic angular frequencies, which correspond to the carrier and upper sideband components in the modulation framework introduced later. In the subsequent sections, we equivalently denote $\omega_1 = \omega_u$ and $\omega_2 = \omega_u + \omega_a$ for consistency with the modulation representation.

The acoustic pressure at a field point $\mathbf{r} = (x, y, z)$ is then obtained as

$$p_a(\mathbf{r}) = -i\rho_0\omega_a \iiint_V q(\mathbf{r}_v) g(\mathbf{r}, \mathbf{r}_v; \omega_a) d^3\mathbf{r}_v. \quad (5)$$

Equations (4) and (5) indicate that the difference-frequency field originates from the nonlinear interaction of the ultrasonic waves. Under baffled conditions, the resulting beam pattern inherits the directivity of the ultrasonic waves and becomes further sharpened. Therefore, the specific placement of time delays—and the corresponding phase distribution imposed at the aperture, whether in the audio, ultrasonic, carrier, or sideband domain—directly governs the spatial distribution of $p_{1,2}(\mathbf{r}_v)$, and hence the audible field $p_a(\mathbf{r})$.

B. Efficient Modeling of Phased-Array PALs Using the k -Space Method

Directly evaluating the Rayleigh integral (2) and the subsequent volume convolution (5) is computationally prohibitive. To avoid this cost, we work in the frequency domain: the ultrasonic primary waves are obtained via the ASA from the aperture velocity, and the audible field is evaluated in k -space by multiplying the virtual-source spectrum with the Green’s function; both fields are finally recovered by inverse FFT. This replaces the Rayleigh and volume integrals with FFT-based spectral multiplications. A more detailed quantitative evaluation of the computational efficiency between the k -space approach and the direct integration method can be found in Ref. [26].

On the baffled aperture S , the source is represented by the normal surface velocity amplitude $v_{i,z}(\mathbf{r}_s)$ at frequency f_i ($i = 1, 2$), where $\mathbf{r}_s = (x_s, y_s, 0)$. For a continuous profile,

$$v_{i,z}(\mathbf{r}_s) = v_{0,i}(\mathbf{r}_s) e^{i\Phi_i(\mathbf{r}_s)}. \quad (6)$$

Here, the pair $(v_{0,i}(\mathbf{r}_s), \Phi_i(\mathbf{r}_s))$ encodes the phased-aperture control: Φ_i specifies the phase distribution for steering or focusing, while $v_{0,i}$ implements amplitude taper for sidelobe control. For example, steering the beam in the x - z plane toward the angle θ_0 can be implemented by the linear phase profile $\Phi_i(\mathbf{r}_s) = k_i x_s \sin \theta_0$. The corresponding angular spectrum is obtained by the 2-D Fourier transform over S , where (k_x, k_y) denote the transverse spatial-frequency variables: [29], [30]:

$$V_i(k_x, k_y) = \iint_S v_{i,z}(\mathbf{r}_s) e^{-i(k_x x_s + k_y y_s)} d^2\mathbf{r}_s. \quad (7)$$

With $V_i(k_x, k_y)$ known, we follow the ASA: multiply $V_i(k_x, k_y)$ by the free-space transfer function in k -space and apply an inverse 2-D Fourier transform to obtain $p_i(\mathbf{r})$ on the plane z (in practice evaluated at $\mathbf{r}_v$).

When the aperture consists of a finite set of M transducers indexed by $n = 0, 1, \dots, M-1$, each element has a finite aperture. The array is therefore modeled as

$$v_{i,z}(\mathbf{r}_s) = \sum_{n=0}^{M-1} A_{i,n} e^{i\phi_{i,n}} a_n(\mathbf{r}_s - \mathbf{r}_n), \quad \mathbf{r}_n = (x_n, y_n, 0), \quad (8)$$

where $a_n(\cdot)$ is the element-aperture function, defined in the local coordinates $\boldsymbol{\rho} = \mathbf{r}_s - \mathbf{r}_n$ to indicate the active region of the n -th element (e.g., for a circular piston of radius a , $a_n(\boldsymbol{\rho}) = \Theta(a - \|\boldsymbol{\rho}\|)$ with Θ the Heaviside step function).

The corresponding discrete angular spectrum reads

$$V_i(k_x, k_y) = \sum_{n=0}^{M-1} A_{i,n} e^{i\phi_{i,n}} A_n(k_x, k_y) e^{-i(k_x x_n + k_y y_n)}, \quad (9)$$

with the element factor defined in (10):

$$A_n(k_x, k_y) = \iint a_n(\boldsymbol{\rho}) e^{-i(k_x \rho_x + k_y \rho_y)} d^2\boldsymbol{\rho}. \quad (10)$$

If all elements share the same aperture, i.e., $a_n \equiv a_e$, then $A_n(k_x, k_y) \equiv A_e(k_x, k_y)$. Substituting this into (9), we obtain

$$V_i(k_x, k_y) = A_e(k_x, k_y) \sum_{n=0}^{M-1} A_{i,n} e^{i\phi_{i,n}} e^{-i(k_x x_n + k_y y_n)}. \quad (11)$$

Equations (6)–(7) describe the general continuous surface-velocity formulation within the k -space framework, and the discrete array formulation in (8)–(11) represents its sampled implementation.

Once the ultrasonic fields are obtained, the audio sound field is computed uniformly in the k -space domain. According to the quasi-linear theory, the volume convolution of the virtual source distribution $q(\mathbf{r}_v)$ in (5) can be recast as

$$P_a(\mathbf{k}) = -i\rho_0\omega_a Q(\mathbf{k}) G_a(\mathbf{k}), \quad (12)$$

where $\mathbf{k} = (k_x, k_y, k_z)$ denotes the three-dimensional spatial-frequency vector, $Q(\mathbf{k})$ is the 3D Fourier transform of $q(\mathbf{r}_v)$, and $G_a(\mathbf{k})$ is the Green’s function in the spatial-frequency domain [26]. The audible pressure field $p_a(\mathbf{r})$ is then recovered efficiently via inverse 3D FFT.

It is noted that the above formulation follows the quasi-linear approximation commonly adopted in PAL analysis [26], which assumes moderate ultrasonic excitation levels and weak nonlinearity. At very high excitation levels (> 130 dB for ultrasound), the acoustic nonlinearity may become too strong and fall outside the scope of the present model.

This unified k -space description—consistent with the quasi-linear formulation and accommodating both continuous and finite-size sampled apertures—serves as the basis for the five delay-placement strategies introduced next.

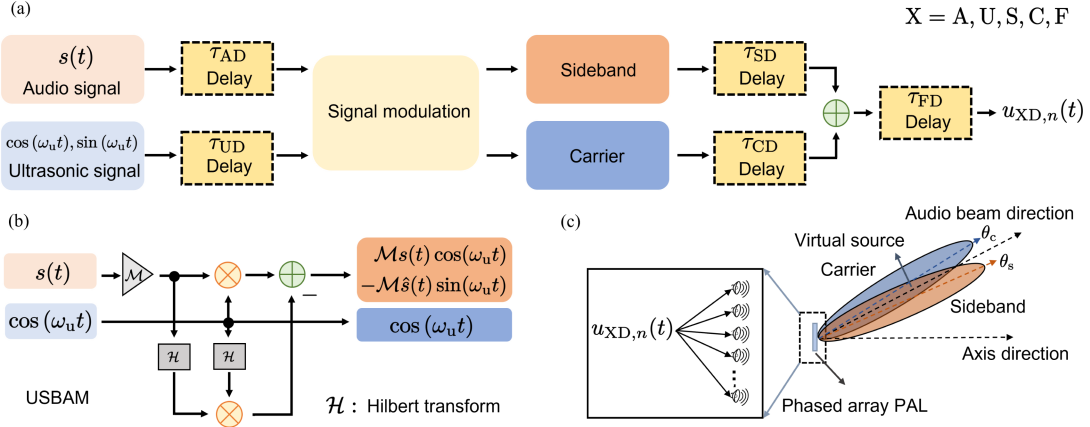


Fig. 2. Illustration of delay-placement strategies for phased-array PALs. (a) Five representative delay-placement strategies, where time delays can be applied to the audio signal (AD), the ultrasonic signal (UD), the sideband (SD), the carrier (CD), or the full band (FD). (b) Signal modulation using USBAM, where the audio signal $s(t)$ and its Hilbert transform $\hat{s}(t)$ generate the upper sideband together with the carrier $\cos(\omega_u t)$. (c) Conceptual explanation of the UD strategy based on the quasi-linear solution. Under pure-tone excitation, the delayed ultrasonic signal causes the carrier and sideband to be steered with slightly different phase gradients, which determines the virtual-source distribution and thus the direction of the audio beam.

C. Five Time-Delay Strategies

To provide a unified framework for analyzing different delay-placement strategies, we adopt the single-sideband amplitude modulation (SSBAM) representation with residual carrier. As illustrated in Fig. 2, the upper sideband (USBAM) configuration is used for clarity. Compared with conventional double-sideband amplitude modulation (DSBAM), SSBAM suppresses redundant cross-terms in the quasi-linear source and retains only the interaction between the carrier and one sideband. This representation clarifies the correspondence between the applied time delays and the resulting ultrasonic phase gradient that governs the audible difference-frequency beam.

Let the broadband audio signal be denoted by $s(t)$, and its Hilbert transform $\mathcal{H}\{s(t)\} = \hat{s}(t)$. Using the in-phase quadrature carrier components, $\cos(\omega_u t)$ and $\sin(\omega_u t)$, the USBAM signal with residual carrier can be expressed as

$$u(t) = \cos(\omega_u t) + \mathcal{M}\{s(t)\cos(\omega_u t) - \hat{s}(t)\sin(\omega_u t)\}, \quad (13)$$

where $\mathcal{M}$ is the modulation index that scales the upper sideband; with peak-normalized $s(t)$, $0 \leq \mathcal{M} \leq 1$ avoids over-modulation [31]. This expression consists of the residual carrier plus a single upper sideband, and is valid for broadband modulation.

The USBAM with residual carrier can be represented as a two-input/two-output (2×2 -port) signal flow structure. The inputs are the broadband audio signal $s(t)$ and the ultrasonic signal $\cos(\omega_u t)$, while the outputs are the residual carrier and the upper sideband.

As shown in Fig. 2(a), the 2×2 -port framework is used to localize where delays are applied along the signal flow. In this representation we define five delay-placement strategies: (i) **AD**—apply a delay on the audio input only; (ii) **UD**—apply a delay on the ultrasonic input only; (iii) **CD**—insert a delay on the carrier branch after modulation; (iv) **SD**—insert a delay on the sideband branch after modulation; and (v) **FD**—insert a delay on the entire modulated signal. The delays associated with

these strategies are denoted by τ_{AD} , τ_{UD} , τ_{CD} , τ_{SD} , and τ_{FD} . The dashed boxes in Fig. 2(a) indicate that the delay is inserted at only one location for a given strategy, i.e., the delay placements are mutually exclusive.

- **AD (audio-only delay):**

$$u_{AD}(t) = \cos(\omega_u t) + \mathcal{M}\{s(t - \tau_{AD})\cos(\omega_u t) - \hat{s}(t - \tau_{AD})\sin(\omega_u t)\}. \quad (14)$$

- **UD (ultrasound-only delay):**

$$u_{UD}(t) = \cos[\omega_u(t - \tau_{UD})] + \mathcal{M}\{s(t)\cos[\omega_u(t - \tau_{UD})] - \hat{s}(t)\sin[\omega_u(t - \tau_{UD})]\}. \quad (15)$$

- **CD (carrier-only delay):**

$$u_{CD}(t) = \cos[\omega_u(t - \tau_{CD})] + \mathcal{M}\{s(t)\cos(\omega_u t) - \hat{s}(t)\sin(\omega_u t)\}. \quad (16)$$

- **SD (sideband-only delay):**

$$u_{SD}(t) = \cos(\omega_u t) + \mathcal{M}\{s(t - \tau_{SD})\cos[\omega_u(t - \tau_{SD})] - \hat{s}(t - \tau_{SD})\sin[\omega_u(t - \tau_{SD})]\}. \quad (17)$$

- **FD (full-band delay):**

$$u_{FD}(t) = \cos[\omega_u(t - \tau_{FD})] + \mathcal{M}\{s(t - \tau_{FD})\cos[\omega_u(t - \tau_{FD})] - \hat{s}(t - \tau_{FD})\sin[\omega_u(t - \tau_{FD})]\}. \quad (18)$$

D. Phased-Array Beam Steering

Consider an N -element uniform linear array (ULA) with inter-element spacing d and sound speed c_0 . For a far-field steering angle θ_0 (relative to broadside), the time-delay law is

$$\tau_n = n \Delta \tau = \frac{nd}{c_0} \sin \theta_0, \quad n = 0, \dots, N-1. \quad (19)$$

With sampling period $T_s = 1/f_s$, where f_s is the sampling rate of the implementation system, integer-sample delays quantize the attainable angles to

$$\theta_m = \arcsin\left(\frac{c_0}{d} m T_s\right), \quad m \in \mathbb{N}, \quad (20)$$

revealing the sampling-limited steering resolution.

It should be noted that fractional delay filters can be employed to achieve sub-sample delay resolution and thus relax the integer-sampling constraint implied by (20). However, in high-sampling-rate multi-channel PAL systems, implementing broadband fractional delays for each channel significantly increases computational and hardware complexity, particularly under real-time constraints.

E. Comparison of Delay Strategies

To facilitate comparison, we examine the case of a single-tone audio input $s(t) = \cos(\omega_a t)$ with modulation index $\mathcal{M} = 1$. Under this condition, the modulated signal in (13) reduces to

$$u(t) = \cos(\omega_u t) + \cos[(\omega_u + \omega_a)t]. \quad (21)$$

which consists of the residual carrier and a single spectral component at $\omega_u + \omega_a$.

Under the quasi-linear framework, the carrier at ω_u and the sideband at $\omega_u + \omega_a$ act as two primary ultrasonic fields. Applying time delays at different stages is thus equivalent to steering these two components with distinct phase profiles. The degree of alignment or mismatch between their steering directions governs the effective virtual-source distribution, and thereby explains the relative effectiveness of the five delay-placement strategies, as illustrated in Fig. 2(c).

Under the time-delay law, these two components may experience different steering angles depending on where the delay is applied. Let the nominal steering angle be defined as

$$\sin \theta_0 = \frac{c_0}{d} \Delta\tau, \quad (22)$$

then the effective steering angles of the carrier (θ_c) and sideband (θ_s), together with the resulting beamforming performance, can be derived for each strategy as follows:

- **AD (audio-only delay):**

$$\begin{aligned} u_{AD}(t) &= \cos(\omega_u t) \\ &+ \cos[(\omega_u + \omega_a)t - \omega_a \tau_{AD}]. \end{aligned} \quad (23)$$

$$\sin \theta_c = 0, \quad \sin \theta_s = \frac{\omega_a}{\omega_u + \omega_a} \sin \theta_0 \xrightarrow{\omega_a \ll \omega_u} 0.$$

This strategy applies delay only at the audio frequency, so the ultrasonic waves remain nearly coaxial. The audible beam cannot be effectively steered and the main lobe broadens.

- **UD (ultrasound-only delay):**

$$\begin{aligned} u_{UD}(t) &= \cos[\omega_u(t - \tau_{UD})] \\ &+ \cos[(\omega_u + \omega_a)t - \omega_u \tau_{UD}]. \end{aligned} \quad (24)$$

$$\sin \theta_c = \sin \theta_0, \quad \sin \theta_s = \frac{\omega_u}{\omega_u + \omega_a} \sin \theta_0 \xrightarrow{\omega_a \ll \omega_u} \sin \theta_0.$$

Both carrier and sideband are approximately co-steered, with a small mismatch due to frequency difference. Steering is effective with high efficiency, though not fully optimal compared to FD.

- **CD (carrier-only delay):**

$$\begin{aligned} u_{CD}(t) &= \cos[\omega_u(t - \tau_{CD})] \\ &+ \cos[(\omega_u + \omega_a)t]. \end{aligned} \quad (25)$$

$$\sin \theta_c = \sin \theta_0, \quad \sin \theta_s = 0.$$

Only the carrier is steered while the sideband remains broadside. This walk-off reduces virtual-source overlap, broadens the main lobe, and introduces asymmetric side-lobes into the beam pattern.

- **SD (sideband-only delay):**

$$\begin{aligned} u_{SD}(t) &= \cos(\omega_u t) \\ &+ \cos[(\omega_u + \omega_a)(t - \tau_{SD})]. \end{aligned} \quad (26)$$

$$\sin \theta_c = 0, \quad \sin \theta_s = \sin \theta_0.$$

Only the sideband is steered, leading to similar walk-off as CD. Performance is intermediate, and the shorter wavelength of the sideband makes it sensitive to spatial aliasing.

- **FD (full-band delay):**

$$\begin{aligned} u_{FD}(t) &= \cos[\omega_u(t - \tau_{FD})] \\ &+ \cos[(\omega_u + \omega_a)(t - \tau_{FD})]. \end{aligned} \quad (27)$$

$$\sin \theta_c = \sin \theta_0, \quad \sin \theta_s = \sin \theta_0.$$

Carrier and sideband are aligned with matched delays. This achieves optimal steering and beam quality, at the expense of higher implementation cost due to wideband fractional delays.

The beam behavior of the above five distinct strategies can be interpreted in terms of the relative directional alignment between the carrier and the sideband. Since the audible field originates from their nonlinear interaction, any angular mismatch between these two ultrasonic components leads to a spatially spread virtual-source distribution.

This mechanism can be clarified by comparing each strategy with the ideal FD case in (27). In (27), the ultrasonic signal consists of two cosine components corresponding to the carrier and the sideband. Both components contain the same delay τ_{FD} applied to their respective frequencies, which steers them toward the same direction. This directional alignment provides the ideal condition for generating a well-formed audio beam. In the CD and SD strategies (25) and (26), only one cosine component contains the delay term. Consequently, either the carrier or the sideband is steered while the other remains broadside, producing a strong directional mismatch between the two components. In the UD strategy (24), the carrier contains the target delay term, while the sideband exhibits a slightly different phase term due to the modulation structure. As a result, the two components are nearly co-steered but with a small angular mismatch. In the AD strategy (23), the delay is applied only at the audio frequency. The carrier therefore remains unsteered, while the sideband is steered by a very small angle proportional to $\omega_a/(\omega_u + \omega_a)$.

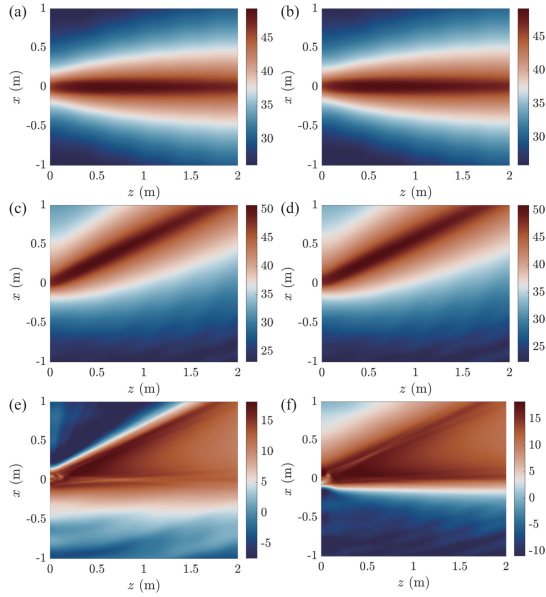


Fig. 3. Audio sound field distributions on the xOz plane ($y = 0$) for $f_a = 500$ Hz and $f_u = 40$ kHz with a beam steering angle of $\theta_0 = -30^\circ$. Subfigures (a–f) correspond to the baseline case ND and strategies AD, UD, FD, CD, and SD, respectively.

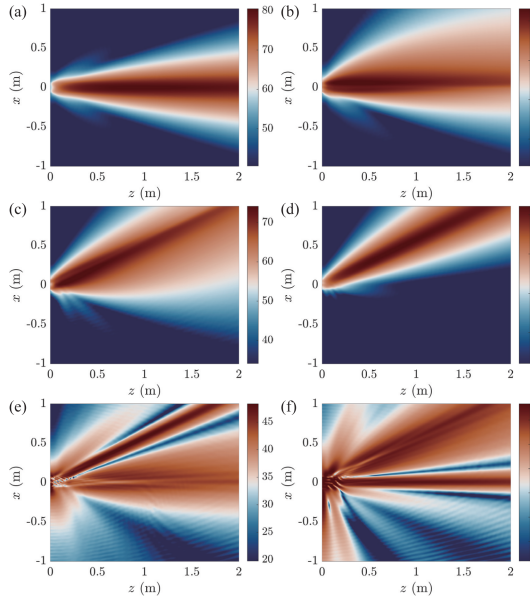


Fig. 4. Audio sound field distributions on the xOz plane ($y = 0$) for $f_a = 4$ kHz and $f_u = 40$ kHz with a beam steering angle of $\theta_0 = -30^\circ$. Subfigures (a–f) correspond to the baseline case ND and strategies AD, UD, FD, CD, and SD, respectively.

Consequently, both components remain nearly aligned with the array axis, leading to a broad and weakly steerable beam.

III. SIMULATIONS

A. Simulation Setup

Numerical simulations were conducted using a k -space method to model the nonlinear self-demodulation of a steered rectangular aperture source. The medium was air with a sound

speed of $c_0 = 343$ m/s, density $\rho_0 = 1.21$ kg/m³, and nonlinearity coefficient $\beta = 1.2$. The environmental conditions were set to a temperature of 20 °C and a relative humidity of 70%. The surface velocity amplitude was $v_0 = 0.121$ m/s, and the target steering angle was $\theta_0 = -30^\circ$.

The source was modeled as an ideal continuous rectangular aperture of size $W \times H$ (with $W = 0.165$ m and $H = 0.109$ m), driven at a uniform normal surface velocity v_0 with a linear steering phase:

$$v_s(x, y; \omega) = v_0 \text{rect}\left(\frac{x}{W}\right) \text{rect}\left(\frac{y}{H}\right) e^{ikx \sin \theta_0}, \quad (28)$$

where $\text{rect}(\xi) = 1$ for $|\xi| \leq 1/2$ and 0 otherwise, and $k = 2\pi f/c_0$ is the wavenumber at frequency f .

Five time-delay strategies were investigated: AD, UD, FD, CD, and SD. A baseline case without delay (ND, No delay) was also included as a reference. For each strategy, the audio sound field distribution was computed on the $y = 0$ cross-section (xOz plane), and the angular response at $r = 1$ m was obtained by sampling the field along a circular arc.

This continuous-source model eliminates the discretization and spatial sampling effects that would otherwise introduce aliasing grating lobes. It therefore provides a theoretical benchmark that highlights the intrinsic influence of delay strategies on beam steering.

B. Audio Sound Performance With Different Delay Strategies

We first presented simulations of a steered rectangular aperture source as a theoretical benchmark at a fixed ultrasonic carrier of $f_u = 40$ kHz. Figures 3 and 4 show the audio sound field distributions on the xOz plane ($y = 0$) at two representative audio frequencies: $f_a = 500$ Hz and $f_a = 4$ kHz.

It is observed that the baseline case ND [Figs. 3(a) and 4(a)] produces a symmetric field without beam steering. Strategy AD [Figs. 3(b) and 4(b)] applies delays only at the audio frequency, leaving the ultrasonic primary waves nearly coaxial, so the resulting audible beam exhibits almost no steering effect. Additionally, the main lobe becomes broader, consistent with the analysis in (14). Strategy FD [Figs. 3(d) and 4(d)] aligns both primary waves with matched delays and achieves the most concentrated beam, as expected from (18). Strategy UD [Figs. 3(c) and 4(c)] produces a well-steered beam comparable to that of strategy FD, but its sidelobes are noticeably wider, reflecting the slight mismatch between the carrier and sideband predicted by (15). Strategies CD [Figs. 3(e) and 4(e)] and SD [Figs. 3(f) and 4(f)] steer only one of the primary waves, in line with (16) and (17); the resulting beams are weaker, broader, and exhibit asymmetric sidelobes.

Comparing configurations $f_a = 500$ Hz (Fig. 3) and $f_a = 4$ kHz (Fig. 4) yields further observations. First, by comparing FD and the baseline case ND we confirm that higher audio frequencies inherently produce narrower main lobes, independent of delay strategies. Second, Strategy AD becomes less effective at low audio frequencies. According to (23), $\sin \theta_s = \frac{\omega_a}{\omega_u + \omega_a} \sin \theta_0$, the ratio $\omega_a/(\omega_u + \omega_a)$ is small when f_a is low. For instance, with $f_u = 40$ kHz it is only about 0.012 at $f_a = 500$ Hz, so the steering angle nearly vanishes and the

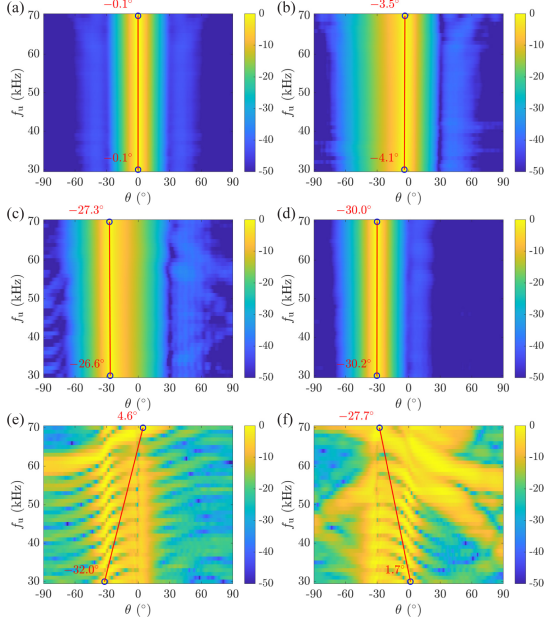


Fig. 5. The frequency-angle maps $L_{\text{norm}}(f_u, \theta)$ with $f_a = 4$ kHz, obtained by stacking the angular responses sampled at $r = 1$ m across the ultrasonic-frequency sweep from 30 kHz to 70 kHz. Results are normalized at each ultrasonic frequency. Subfigures (a–f) correspond to the baseline case ND and strategies AD, UD, FD, CD, and SD, respectively. The red line connects the peak points of $L_{\text{norm}}(f_u, \theta)$ at $f_u = 30$ kHz and $f_u = 70$ kHz, indicating the frequency-dependent walk-off of the main lobe. The target steering angle is -30° .

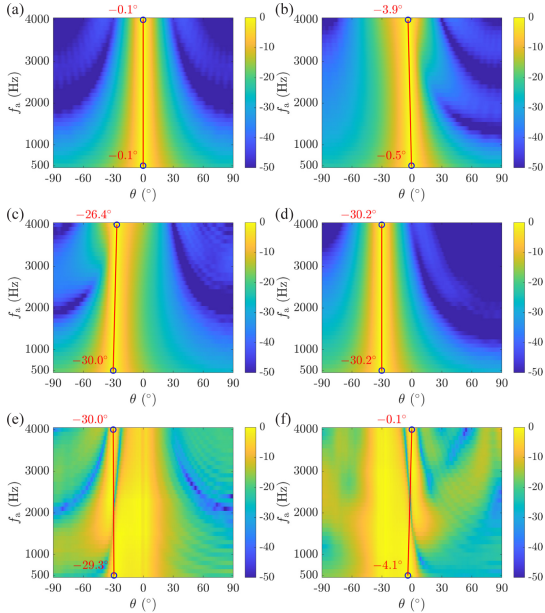


Fig. 6. The frequency-angle maps $L_{\text{norm}}(f_a, \theta)$ with $f_u = 40$ kHz, obtained by stacking the angular responses sampled at $r = 1$ m over the audio-frequency sweep from 500 Hz to 4 kHz. Results are normalized at each audio frequency. Subfigures (a–f) correspond to the baseline case ND and strategies AD, UD, FD, CD, and SD, respectively. The red line connects the peak points of $L_{\text{norm}}(f_a, \theta)$ at $f_a = 500$ Hz and $f_a = 4$ kHz, highlighting the frequency-dependent deviation of the main-lobe direction from the target steering angle. The target steering angle is -30° .

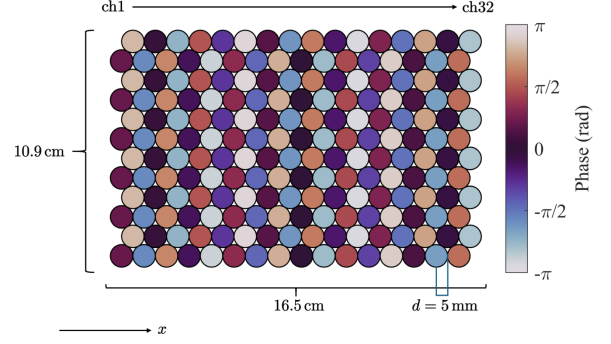


Fig. 7. Layout of the 32×6 PAL prototype used in the experiments at $f_u = 40$ kHz and $f_a = 500$ Hz. The array aperture measures $16.5 \text{ cm} \times 10.9 \text{ cm}$ with a channel pitch of $d = 5$ mm. Each circle represents one ultrasonic transducer element, and the color indicates the applied phase (rad) at a steering angle of -30° .

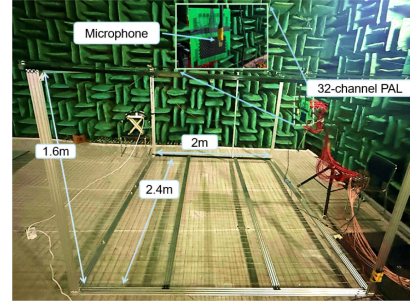


Fig. 8. Experimental setup showing the PAL system, microphone, and the motorized sliding rail ($2.4 \text{ m} \times 2.0 \text{ m} \times 1.6 \text{ m}$) used for sound field scanning in the anechoic chamber.

beam remains close to broadside. Even at $f_a = 4$ kHz the ratio is still only about 0.09, giving limited steering performance. Third, strategy UD performs well at low audio frequencies but deteriorates at higher ones. As (24) shows, $\sin \theta_s = \frac{\omega_u}{\omega_u + \omega_a} \sin \theta_0$, the ratio is close to one when f_a is small. At $f_a = 500$ Hz it reaches 0.99 for $f_u = 40$ kHz, so the main lobe almost coincides with θ_0 . When $f_a = 4$ kHz the ratio decreases to 0.91, causing the beam to deviate slightly from θ_0 and producing stronger sidelobes. Finally, CD and SD remain the weakest strategies across all cases; at higher audio frequencies, their beams suffer more severely from sidelobe leakage.

C. Effects of Frequencies

To investigate the effects of frequency variation, we constructed frequency-angle maps by sweeping either the ultrasonic frequency or the audio frequency while keeping the other fixed. For each tested frequency, the audible field was computed from the ultrasonic primary waves under USBAM with residual carrier error.

The angular response at $r = 1$ m was obtained by sampling the field along a circular arc and converted to sound pressure level (SPL) as

$$L(f, \theta) = 20 \log_{10} \left(\frac{p_{\text{rms}}(f, \theta)}{p_{\text{ref}}} \right), \quad p_{\text{ref}} = 20 \mu\text{Pa}, \quad (29)$$

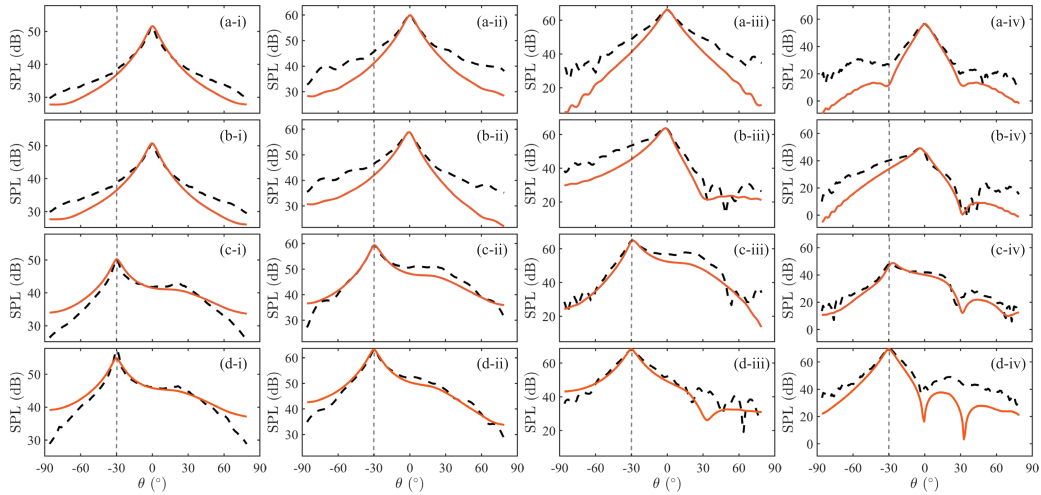


Fig. 9. Experimental and simulated angular responses of the PAL under different delay strategies at a propagation distance of 1 m. Rows (a–d) correspond to the baseline case ND and strategies AD, UD, and FD, respectively. Columns (i–iv) show results for $f_a = 500$ Hz, 1 kHz, 2 kHz, and 4 kHz. Black dashed lines: measurements; red solid lines: k -space simulations; gray dashed line: target steering angle (-30°).

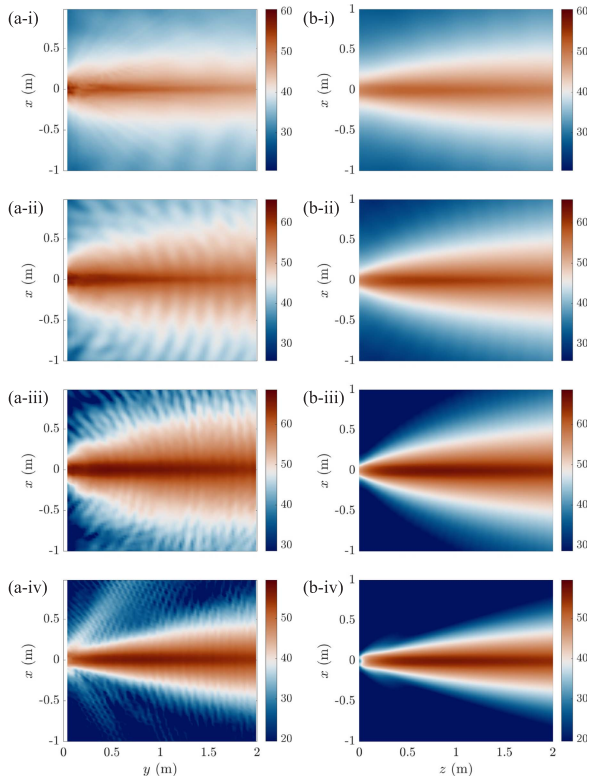


Fig. 10. Experimental and simulated SPL distributions of the PAL under the baseline case ND in the xOz -plane. Columns (a–b): (a) experimental; (b) k -space simulation. Rows (i–iv): $f_a = 500$ Hz, 1 kHz, 2 kHz, and 4 kHz.

where f denotes the swept frequency variable, i.e., either f_u or f_a , and $p_{\text{rms}}(f, \theta) = |p(f, \theta)|/\sqrt{2}$ denotes the root-mean-square (RMS) pressure converted from the peak pressure amplitude $p(f, \theta)$.

Next, to remove the absolute amplitude dependence, the SPL was normalized with respect to its angular maximum:

$$L_{\text{norm}}(f, \theta) = L(f, \theta) - \max_{\theta} L(f, \theta). \quad (30)$$

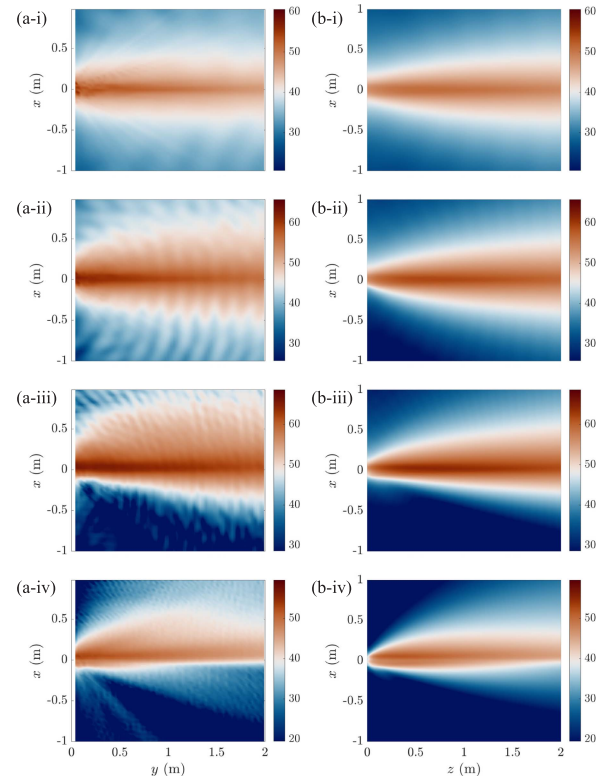


Fig. 11. Experimental and simulated SPL distributions of the PAL under strategy AD in the xOz -plane. Columns (a–b): (a) experimental; (b) k -space simulation. Rows (i–iv): $f_a = 500$ Hz, 1 kHz, 2 kHz, and 4 kHz.

Figures 5 and 6 summarize the frequency-angle maps for the baseline case ND and the five delay strategies under ultrasonic-frequency and audio-frequency sweeps, respectively. In both figures, the red line connects the peak angles at the two ends of the sweep range to highlight the frequency dependence of the main-lobe direction.

For the baseline case ND, no steering effect is observed, and the main lobe remains near broadside throughout both sweeps.

For strategy AD, the sideband angle follows (23), $\sin \theta_s = \frac{\omega_a}{\omega_u + \omega_a} \sin \theta_0$, so the beam remains close to broadside because the ratio is always small. Accordingly, only weak steering is observed in both maps: under the ultrasonic-frequency sweep, the peak shifts slightly from -4.1° to -3.5° , whereas under the audio-frequency sweep it moves from -0.5° to -3.9° .

For strategy UD, the sideband angle follows (24), $\sin \theta_s = \frac{\omega_u}{\omega_u + \omega_a} \sin \theta_0$. Thus, when f_u increases, the ratio approaches unity and the beam converges toward the target direction, as seen in Fig. 5(c), where the peak shifts from -26.6° to -27.3° . By contrast, when f_a increases, the ratio decreases and the beam moves slightly away from the target, consistent with Fig. 6(c), where the peak changes from -30.0° to -26.4° .

For strategies CD and SD, the carrier and sidebands are not directionally aligned, so their beam behavior is degraded and cannot be characterized by a single physically meaningful steering angle.

In addition, Fig. 6 shows that the beamwidth variation with f_a is strategy dependent. For ND and FD, the main lobe generally becomes narrower as f_a increases, consistent with the increase in acoustic directivity at higher audio frequencies. By contrast, for AD and UD, the main lobe first narrows at low and medium audio frequencies, but becomes broader again toward the high-frequency end. This reversal indicates that, although the intrinsic directivity tends to increase with frequency, the growing mismatch between the two ultrasonic components gradually dominates the beam pattern and leads to renewed broadening of the audible beam.

Taken together, the simulations show that strategy AD scarcely steers the audible beam and exhibits a frequency-dependent pointing bias; strategy FD achieves optimal steering by co-aligning the carrier and sideband; strategy UD delivers performance close to FD, with only a small under-steering, while being much simpler to implement (pure-tone phase shifts, low computational load, and easy channel scaling for real-time control); by contrast, strategies CD and SD steer only one primary wave and therefore produce weaker, broader beams.

IV. EXPERIMENTS

A. Experiment Setup

To validate the theoretical predictions and simulation results, a compact multi-channel PAL prototype was fabricated, with the sketch shown in Fig. 7. The prototype consists of 192 circular ultrasonic transducers MA40S4S (Murata Manufacturing Co., Ltd., Kyoto, Japan), each with a resonance frequency of 40 kHz and a radius of 5 mm. The array is organized into 32 channels, with each channel comprising 6 transducer elements arranged in a vertical stack. The channel spacing along the horizontal direction is 5 mm, forming a dense 32×6 configuration. Correspondingly, the same array geometry was implemented in the numerical simulations in this section using the k -space method. Unlike the continuous-aperture benchmark in Section III, the present model reintroduces the discrete array implementation by representing the source as finite-aperture elements located at the measured transducer positions. Since the k -space framework accommodates both continuous and discretized source

distributions, it enables direct comparison with the experimental results while providing a consistent numerical tool for examining practical discretization effects, including spatial sampling, finite element spacing, and array-induced artifacts. This bridges the idealized benchmark analysis in Section III and the practical array validation in Section IV. Each channel is driven by a power amplifier TPA3116D2 (Texas Instruments Inc., Dallas, TX, USA), and the delay for each channel can be independently controlled to realize phased-array operation. The multi-channel delayed signals were generated in MATLAB R2024a (MathWorks Inc., Natick, MA, USA) and fed to the array through an audio interface Orion32+ Gen 3 (Antelope Audio Ltd., Sofia, Bulgaria).

As shown in Fig. 8, the experiments were carried out in the anechoic chamber at Nanjing University, with dimensions of $11.4 \text{ m} \times 7.8 \text{ m} \times 6.7 \text{ m}$ (height), under environmental conditions of 50% relative humidity and 27°C temperature. Sound pressure was measured using a CHZ-221 condenser microphone (Hangzhou Acoustic Sensing Technology Co., Ltd., Hangzhou, China), and the signals were conditioned and analyzed with a PULSE analyzer Type 3160 (Brüel & Kjær A/S, Nærum, Denmark). To mitigate spurious sound generated by strong ultrasonic waves, the microphone was covered with a thin plastic film.

The rail spans $2.4 \text{ m} \times 2.0 \text{ m} \times 1.6 \text{ m}$, and measurements were conducted on a plane with dimensions of $2.0 \text{ m} \times 2.0 \text{ m}$ and a 30 mm stepsize. The radiating surface of the PAL was oriented perpendicular to the measurement plane. The microphone was positioned at the same height as the array center and was scanned across the measurement area by a three-axis motorized translation stage, which was controlled by a personal computer.

B. Experiment Results

1) *Angular Patterns*: Based on the setup described in Section IV-A, the angular response performance of different delay-placement strategies was evaluated at a propagation distance of 1 m. Three representative strategies (AD, UD, and FD) were implemented on the 32×6 PAL prototype, together with the baseline case ND. The CD and SD strategies were excluded from experiments, as theoretical analysis predicts severe carrier-sideband misalignment and poor beam formation. The measured angular responses (black dashed) are compared with k -space simulations (red solid) in Fig. 9. Overall, the measurements show great agreement with the simulations, validating both the experimental system and the modeling approach. Minor discrepancies, such as elevated sidelobe levels and small peak deviations, are attributed to non-uniform transducer response, imperfect alignments, and measurement uncertainties.

In the baseline case ND [Fig. 9(a)], the audio beam remains nearly coaxial with the array, serving as a reference for assessing the impact of delay placement. Under strategy AD [Fig. 9(b)], stable beam steering cannot be achieved. The phase mismatch between the carrier and sidebands leads to broadened main lobes and spurious leakage, which becomes more pronounced as the audio frequency increases. With strategy UD [Fig. 9(c)], steering becomes effective across the frequency band. The low-frequency response closely resembles that of FD, although

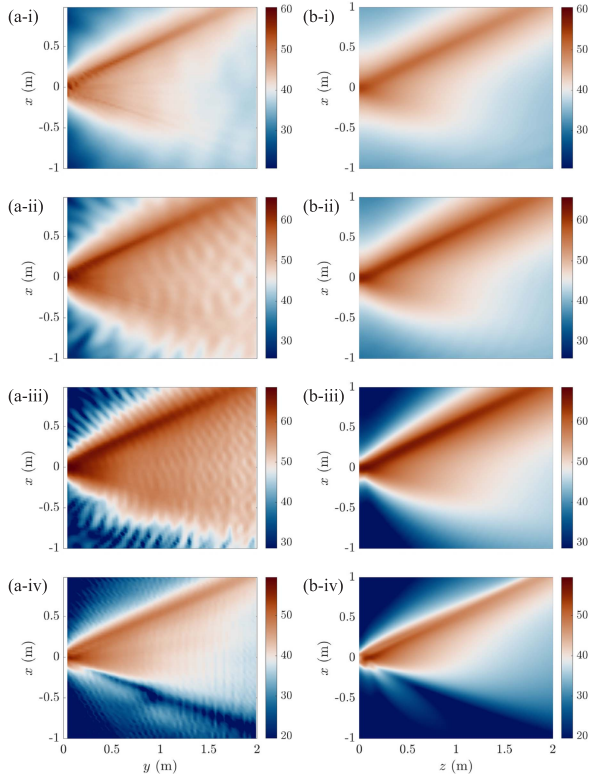


Fig. 12. Experimental and simulated SPL distributions of the PAL under strategy UD in the xOz -plane. Columns (a–b): (a) experimental; (b) k -space simulation. Rows (i–iv): $f_a = 500$ Hz, 1 kHz, 2 kHz, and 4 kHz.

at higher frequencies the main-lobe contrast weakens and sidelobes become more noticeable. Finally, strategy FD [Fig. 9(d)] provides the most accurate and robust steering performance, exhibiting well-defined main lobes, strong sidelobe suppression, and close agreement with simulations over the entire band.

2) *Two-Dimensional (2D) Audio Sound Field Distributions:* Both simulated and measured 2D audio sound field distributions in the xOz -plane are shown in Figs. 10–13 for the baseline case ND and the strategies AD, UD, and FD, respectively. In the baseline case ND (Fig. 10), the directional audio beam is observed, a distinct property of the PAL without beam steering. Strategy AD (Fig. 11) produces weak and unstable beam steering, with broadened lobes and asymmetric leakage at higher audio frequencies. Strategy UD (Fig. 12) produces clear steering effects, although higher audio frequencies reveal slightly wider beams and stronger sidelobes compared with strategy FD. Strategy FD (Fig. 13) produces well-steered beams that closely match the simulations.

In the measured maps, weak periodic ripple patterns can be observed along the propagation direction. These ripples are attributed to environmental reflections in the measurement space that interfere with the primary beam. The ripple spacing decreases with increasing audio frequency and is approximately consistent with the interference distance $\lambda_a/2 = c/(2f_a)$, which is characteristic of weak multipath interference.

In summary, the measured 2D field distributions corroborate the angular-response observations: strategy AD fails to generate

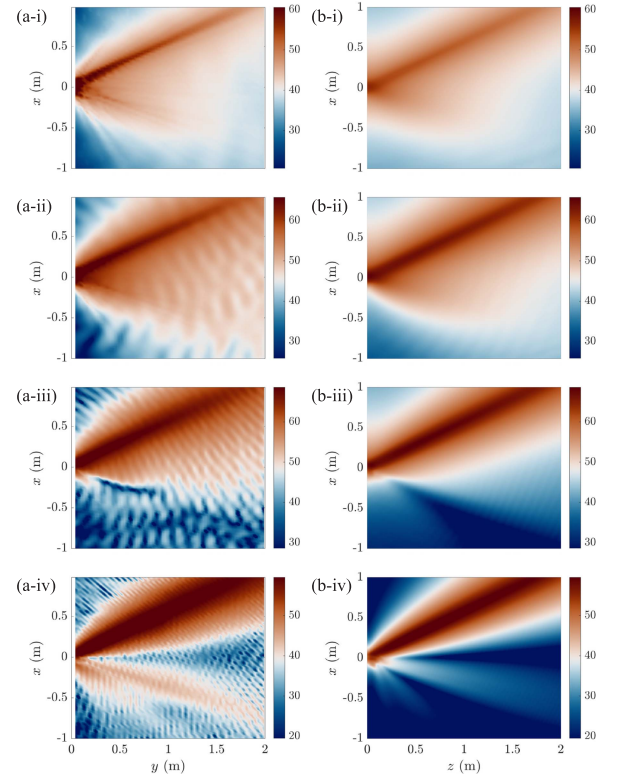


Fig. 13. Experimental and simulated SPL distributions of the PAL under strategy FD in the xOz -plane. Columns (a–b): (a) experimental; (b) k -space simulation. Rows (i–iv): $f_a = 500$ Hz, 1 kHz, 2 kHz, and 4 kHz.

an effective steered beam, strategy UD provides a near-optimal compromise with mild high-frequency degradation, and strategy FD achieves the most stable and concentrated beamforming. Together, these results confirm that delay placement constitutes a critical new control dimension in phased-array PALs.

V. CONCLUSIONS AND DISCUSSION

This work presented a systematic comparative analysis of five delay-placement strategies (AD, UD, FD, CD, and SD) for phased-array PALs within a unified quasi-linear framework. Although USBAM was adopted for clarity, the conclusions extend to other amplitude-modulation variants, since steering is fundamentally governed by the relative phasing between the carrier and sidebands rather than by the specific modulation scheme. By explicitly linking delay placement to carrier-sideband phase alignment, the proposed framework clarifies the steering mechanisms and the corresponding virtual-source distribution in k -space.

Comprehensive k -space simulations together with measurements on a 32×6 prototype demonstrated consistent behavior across strategies. FD achieves the most accurate beam steering by co-steering both ultrasonic components. UD attains nearly comparable performance while requiring only single-frequency phase shifts, thereby providing a favorable accuracy-complexity trade-off. AD fails to generate effective steering due to the absence of an ultrasonic phase gradient, whereas CD and SD

suffer from carrier-sideband walk-off that degrades beam quality. Continuous-aperture simulations established a theoretical benchmark, and discrete-array results confirmed the analysis under realistic sampling and spacing constraints. A key implication is that UD combines hardware efficiency with near-optimal steering performance, making it particularly attractive for real-time PAL implementations.

From a practical design perspective, selecting a delay-placement strategy requires balancing steering accuracy, implementation complexity, and achievable sound pressure level. In the experimental architecture of Section IV-A, FD necessitates per-channel broadband fractional delay filtering at ultrasonic sampling rates, increasing computational load and memory usage in FPGA-based real-time systems. In contrast, UD reduces the operation to single-frequency phase shifts of the ultrasonic carrier, which can be implemented efficiently with minimal hardware overhead [32]. In systems where independent full-band digital processing is already available per channel (e.g., via multi-channel audio interfaces), the relative implementation advantage of UD may be less pronounced. However, in typical PAL hardware operating at high ultrasonic sampling rates with constrained real-time resources, UD offers a practical compromise by retaining near-optimal steering performance while significantly reducing implementation complexity. Moreover, although increasing ω_u may theoretically improve alignment under $\omega_a \ll \omega_u$, practical limits such as atmospheric absorption, transducer efficiency, and nonlinear conversion efficiency constrain the achievable sound pressure level [33]. Effective PAL design therefore requires jointly considering steering performance, hardware resources, and physical limitations.

Future research directions include extending the comparative framework to broadband audio signals and perceptually motivated evaluation metrics, as well as exploring joint optimization of delay placement with amplitude shading and near-field focusing. Overall, the proposed framework provides both theoretical clarity and practical design guidance, and establishes delay placement as a distinct and effective control dimension in phased-array PALs.

REFERENCES

- [1] M. Brandstein and D. Ward, *Microphone Arrays: Signal Processing Techniques and Applications*. Berlin, Germany: Springer, 2001.
- [2] B. D. V. Veen and K. M. Buckley, "Beamforming: A versatile approach to spatial filtering," *IEEE ASSP Mag.*, vol. 5, no. 2, pp. 4–24, Apr. 1988.
- [3] Y. Li, X. Jiang, B. Liang, J.-C. Cheng, and L. Zhang, "Metascreen-based acoustic passive phased array," *Phys. Rev. Appl.*, vol. 4, no. 2, 2015, Art. no. 024003.
- [4] J. Schmerr and W. Lester, "Fundamentals of ultrasonic phased arrays," *Modern Phys. Lett. B*, vol. 22, no. 11, pp. 917–921, 2008.
- [5] Y. Ochiai, T. Hoshi, and J. Rekimoto, "Three-dimensional mid-air acoustic manipulation by ultrasonic phased arrays," *PLoS One*, vol. 9, no. 5, 2014, Art. no. e97590.
- [6] P. J. Westervelt, "Parametric acoustic array," *J. Acoust. Soc. Amer.*, vol. 35, no. 4, pp. 535–537, 1963.
- [7] W.-S. Gan, J. Yang, and T. Kamakura, "A review of parametric acoustic array in air," *Appl. Acoust.*, vol. 73, no. 12, pp. 1211–1219, 2012.
- [8] J.-X. Zhong, "Parametric array loudspeakers and applications in active noise control," Ph.D. dissertation, Univ. Technol. Sydney, Sydney, NSW, Australia, 2022.
- [9] J.-X. Zhong, R. Kirby, and X. Qiu, "The near field, westervelt far field, and inverse-law far field of the audio sound generated by parametric array loudspeakers," *J. Acoust. Soc. America*, vol. 149, no. 3, pp. 1524–1535, 2021.
- [10] M. Yoneyama, J.-i. Fujimoto, Y. Kawamo, and S. Sasabe, "The audio spotlight: An application of nonlinear interaction of sound waves to a new type of loudspeaker design," *J. Acoust. Soc. Amer.*, vol. 73, no. 5, pp. 1532–1536, 1983.
- [11] W.-S. Gan, J. Yang, K.-S. Tan, and M.-H. Er, "A digital beamsteerer for difference frequency in a parametric array," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 14, no. 3, pp. 1018–1025, May 2006.
- [12] C. Shi, "Investigation of the steerable parametric loudspeaker based on phased array techniques," Ph.D. dissertation, Nanyang Technological Univ., Singapore, 2013.
- [13] F. Fan et al., "A least squares based optimization method for grating lobes suppression in parametric array loudspeakers," *Appl. Acoust.*, vol. 237, 2025, Art. no. 110757.
- [14] Y. Zhu et al., "An enhanced beamsteering algorithm based on MVDR for a multi-channel parametric array loudspeaker array," *J. Sound Vib.*, vol. 595, 2025, Art. no. 118768.
- [15] J.-X. Zhong et al., "Sidelobe suppression for a steerable parametric source using the sparse random array technique," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 3152–3161, 2023.
- [16] T. Zhuang, J.-X. Zhong, G. Ma, F. Niu, and J. Lu, "Modeling a phased array of parametric array loudspeakers using the spherical wave expansion," *IEEE Trans. Audio Speech Lang. Process.*, vol. 33, pp. 309–318, 2025.
- [17] J.-X. Zhong, J. Ji, X. Xia, H. Heo, and Y. Jing, "Audible enclaves crafted by nonlinear self-bending ultrasonic beams," *Proc. Nat. Acad. Sci. USA*, vol. 122, no. 12, 2025, Art. no. e2408975122.
- [18] C. Shi, Y. Kajikawa, and W.-S. Gan, "An overview of directivity control methods of the parametric array loudspeaker," *APSIPA Trans. Signal Inf. Process.*, vol. 3 2014, Art. no. e20.
- [19] D. Olszewski, F. Prasetyo, and K. Linhard, "Steerable highly directional audio beam loudspeaker," in *Proc. Interspeech*, 2005, pp. 137–140.
- [20] J. Yang et al., "Steering of directional sound beams," U.S. Patent 7 146 011, 2006.
- [21] N. Tanaka and M. Tanaka, "Active noise control using a steerable parametric array loudspeaker," *J. Acoust. Soc. Amer.*, vol. 127, no. 6, pp. 3526–3537, 2010.
- [22] S. Wu, M. Wu, C. Huang, and J. Yang, "FPGA-based implementation of steerable parametric loudspeaker using fractional delay filter," *Appl. Acoust.*, vol. 73, no. 12, pp. 1271–1281, 2012.
- [23] A. Marzo, T. Corkett, and B. W. Drinkwater, "Ultrano: An open phased-array system for narrowband airborne ultrasound transmission," *IEEE Trans. Ultrason., Ferroelectr., Freq. Control*, vol. 65, no. 1, pp. 102–111, Jan. 2018.
- [24] X. Cheng, H. Zhao, and X. Zhang, "Design of phase control system for parametric array loudspeakers based on FPGA," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 446, 2020, Art. no. 042084.
- [25] N. Hahn, J. Ahrens, and C. Andersson, "Parametric array using amplitude modulated pulse trains: Experimental evaluation of beamforming and single sideband modulation," in *Proc. 151st Audio Eng. Soc. Conv.*, 2021, pp. 20–26.
- [26] T. Zhuang, L. He, F. Niu, J.-X. Zhong, and J. Lu, "A k -space approach to modeling multi-channel parametric array loudspeaker systems," *J. Acoust. Soc. Amer.*, vol. 158, no. 6, pp. 4651–4661, 2025.
- [27] J. W. Cooley and J. W. Tukey, "An algorithm for the machine calculation of complex Fourier series," *Math. Computation*, vol. 19, no. 90, pp. 297–301, 1965.
- [28] W. M. Gentleman and G. Sande, "Fast Fourier transforms: For fun and profit," in *Proc. Fall Joint Comput. Conf.*, 1966, pp. 563–578.
- [29] E. G. Williams, *Fourier Acoustics: Sound Radiation and Nearfield Acoustical Holography*. San Diego, CA, USA: Elsevier, 1999.
- [30] X. Zeng and R. J. McGough, "Evaluation of the angular spectrum approach for simulations of near-field pressures," *J. Acoust. Soc. Amer.*, vol. 123, no. 1, pp. 68–76, 2008.
- [31] C. Shi and W.-S. Gan, "A preprocessing method to increase high frequency response of a parametric loudspeaker," in *Proc. Asia-Pacific Signal Inf. Process. Assoc. Annu. Summit Conf.*, 2013, pp. 1–5.
- [32] T. Zhuang, M. Li, S. Li, J.-X. Zhong, and J. Lu, "Wideband sound reproduction using a massive multi-channel focusing parametric array loudspeaker system," *J. Acoust. Soc. Amer.*, vol. 158, no. 4, pp. 3339–3347, 2025.
- [33] M. Li, J.-X. Zhong, Y. Jing, and J. Lu, "A note on the audio sound power generated by a parametric array loudspeaker," *J. Acoust. Soc. Amer.*, vol. 154, no. 6, pp. 3899–3905, 2023.